databricks demo / 2.join_demo

(https://databricks.com)

Inner Join

```
%run "../includes/configuration"
```

```
races_df = spark.read.parquet(f"{processed_folder_path}/races").filter("race_year = 2019") \
    .withColumnRenamed("name", "race_name")
```

```
circuits_df = spark.read.parquet(f"{processed_folder_path}/circuits") \
    .filter("circuit_id < 70") \
    .withColumnRenamed("name", "circuit_name")
```

```
display(races_df)
```

Table

	race_id	race_year	round	circuit_id	race_name	ingestion_date	race_timesta
1	1010	2019	1	1	Australian Grand Prix	2024-07-21T09:58:36.897+00:...	2019-03-17T05:
2	1011	2019	2	3	Bahrain Grand Prix	2024-07-21T09:58:36.897+00:...	2019-03-31T15:
3	1012	2019	3	17	Chinese Grand Prix	2024-07-21T09:58:36.897+00:...	2019-04-14T06:
4	1013	2019	4	73	Azerbaijan Grand Prix	2024-07-21T09:58:36.897+00:...	2019-04-28T12:
5	1014	2019	5	4	Spanish Grand Prix	2024-07-21T09:58:36.897+00:...	2019-05-12T13:
6	1015	2019	6	6	Monaco Grand Prix	2024-07-21T09:58:36.897+00:...	2019-05-26T13:
7	1016	2019	7	7	Canadian Grand Prix	2024-07-21T09:58:36.897+00:...	2019-06-09T18:
8	1017	2019	8	34	French Grand Prix	2024-07-21T09:58:36.897+00:...	2019-06-23T13:
9	1018	2019	9	70	Austrian Grand Prix	2024-07-21T09:58:36.897+00:...	2019-06-30T13:
10	1019	2019	10	9	British Grand Prix	2024-07-21T09:58:36.897+00:...	2019-07-14T13:
11	1020	2019	11	10	German Grand Prix	2024-07-21T09:58:36.897+00:...	2019-07-28T13:
12	1021	2019	12	11	Hungarian Grand Prix	2024-07-21T09:58:36.897+00:...	2019-08-04T13:
13	1022	2019	13	13	Belgian Grand Prix	2024-07-21T09:58:36.897+00:...	2019-09-01T13:
14	1023	2019	14	14	Italian Grand Prix	2024-07-21T09:58:36.897+00:...	2019-09-08T13:

21 rows

```
display(circuits_df)
```

Table

	circuit_Id	circuit_ref	circuit_name	location	country	1.2 latitude	1.2 lon
1	1	albert_park	Albert Park Grand Prix Circuit	Melbourne	Australia	-37.8497	
2	2	sepang	Sepang International Circuit	Kuala Lumpur	Malaysia	2.76083	
3	3	bahrain	Bahrain International Circuit	Sakhir	Bahrain	26.0325	
4	4	catalunya	Circuit de Barcelona-Catalunya	Montmeló	Spain	41.57	
5	5	istanbul	Istanbul Park	Istanbul	Turkey	40.9517	
6	6	monaco	Circuit de Monaco	Monte-Carlo	Monaco	43.7347	

7	7	villeneuve	Circuit Gilles Villeneuve	Montreal	Canada	45.5
8	8	magny_cours	Circuit de Nevers Magny-Cours	Magny Cours	France	46.8642
9	9	silverstone	Silverstone Circuit	Silverstone	UK	52.0786
10	10	hockenheimring	Hockenheimring	Hockenheim	Germany	49.3278
11	11	hungaroring	Hungaroring	Budapest	Hungary	47.5789
12	12	valencia	Valencia Street Circuit	Valencia	Spain	39.4589
13	13	spa	Circuit de Spa-Francorchamps	Spa	Belgium	50.4372
14	14	monza	Autodromo Nazionale di Monza	Monza	Italy	45.6156

69 rows

```
race_circuits_df = circuits_df.join(races_df, circuits_df.circuit_Id == races_df.circuit_id, "inner") \
    .select(circuits_df.circuit_name, circuits_df.location, circuits_df.country, races_df.race_name, races_df.round)
```

```
display(race_circuits_df)
```

Table

	circuit_name	location	country	race_name	round
1	Albert Park Grand Prix Circuit	Melbourne	Australia	Australian Grand Prix	1
2	Bahrain International Circuit	Sakhir	Bahrain	Bahrain Grand Prix	2
3	Shanghai International Circuit	Shanghai	China	Chinese Grand Prix	3
4	Circuit de Barcelona-Catalunya	Montmeló	Spain	Spanish Grand Prix	5
5	Circuit de Monaco	Monte-Carlo	Monaco	Monaco Grand Prix	6
6	Circuit Gilles Villeneuve	Montreal	Canada	Canadian Grand Prix	7
7	Circuit Paul Ricard	Le Castellet	France	French Grand Prix	8
8	Silverstone Circuit	Silverstone	UK	British Grand Prix	10
9	Hockenheimring	Hockenheim	Germany	German Grand Prix	11
10	Hungaroring	Budapest	Hungary	Hungarian Grand Prix	12
11	Circuit de Spa-Francorchamps	Spa	Belgium	Belgian Grand Prix	13
12	Autodromo Nazionale di Monza	Monza	Italy	Italian Grand Prix	14
13	Marina Bay Street Circuit	Marina Bay	Singapore	Singapore Grand Prix	15
14	Suzuka Circuit	Suzuka	Japan	Japanese Grand Prix	17
15	Autódromo Hermanos Rodríguez	Mexico City	Mexico	Mexican Grand Prix	18

18 rows

Outer Joins

```
# left_outer_join
race_circuits_df = circuits_df.join(races_df, circuits_df.circuit_Id == races_df.circuit_id, "left") \
    .select(circuits_df.circuit_name, circuits_df.location, circuits_df.country, races_df.race_name, races_df.round)
```

```
display(race_circuits_df)
```

Table

	circuit_name	location	country	race_name	round
1	Albert Park Grand Prix Circuit	Melbourne	Australia	Australian Grand Prix	1
2	Sepang International Circuit	Kuala Lumpur	Malaysia	null	null

3	Bahrain International Circuit	Sakhir	Bahrain	Bahrain Grand Prix	2
4	Circuit de Barcelona-Catalunya	Montmeló	Spain	Spanish Grand Prix	5
5	Istanbul Park	Istanbul	Turkey	null	null
6	Circuit de Monaco	Monte-Carlo	Monaco	Monaco Grand Prix	6
7	Circuit Gilles Villeneuve	Montreal	Canada	Canadian Grand Prix	7
8	Circuit de Nevers Magny-Cours	Magny Cours	France	null	null
9	Silverstone Circuit	Silverstone	UK	British Grand Prix	10
10	Hockenheimring	Hockenheim	Germany	German Grand Prix	11
11	Hungaroring	Budapest	Hungary	Hungarian Grand Prix	12
12	Valencia Street Circuit	Valencia	Spain	null	null
13	Circuit de Spa-Francorchamps	Spa	Belgium	Belgian Grand Prix	13
14	Autodromo Nazionale di Monza	Monza	Italy	Italian Grand Prix	14
15	Marina Bay Street Circuit	Marina Bay	Singapore	Singapore Grand Prix	15

69 rows

```
# right_outer_join
race_circuits_df = circuits_df.join(races_df, circuits_df.circuit_Id == races_df.circuit_id, "right") \
    .select(circuits_df.circuit_name, circuits_df.location, circuits_df.country, races_df.race_name, races_df.round)
```

```
display(race_circuits_df)
```

Table

	circuit_name	location	country	race_name	round
1	Albert Park Grand Prix Circuit	Melbourne	Australia	Australian Grand Prix	1
2	Bahrain International Circuit	Sakhir	Bahrain	Bahrain Grand Prix	2
3	Shanghai International Circuit	Shanghai	China	Chinese Grand Prix	3
4	null	null	null	Azerbaijan Grand Prix	4
5	Circuit de Barcelona-Catalunya	Montmeló	Spain	Spanish Grand Prix	5
6	Circuit de Monaco	Monte-Carlo	Monaco	Monaco Grand Prix	6
7	Circuit Gilles Villeneuve	Montreal	Canada	Canadian Grand Prix	7
8	Circuit Paul Ricard	Le Castellet	France	French Grand Prix	8
9	null	null	null	Austrian Grand Prix	9
10	Silverstone Circuit	Silverstone	UK	British Grand Prix	10
11	Hockenheimring	Hockenheim	Germany	German Grand Prix	11
12	Hungaroring	Budapest	Hungary	Hungarian Grand Prix	12
13	Circuit de Spa-Francorchamps	Spa	Belgium	Belgian Grand Prix	13
14	Autodromo Nazionale di Monza	Monza	Italy	Italian Grand Prix	14
15	Marina Bay Street Circuit	Marina Bay	Singapore	Singapore Grand Prix	15

21 rows

```
# full_outer_join
race_circuits_df = circuits_df.join(races_df, circuits_df.circuit_Id == races_df.circuit_id, "full") \
    .select(circuits_df.circuit_name, circuits_df.location, circuits_df.country, races_df.race_name, races_df.round)
```

```
display(race_circuits_df)
```

Table

	circuit_name	location	country	race_name	round
1	Albert Park Grand Prix Circuit	Melbourne	Australia	Australian Grand Prix	1
2	Sepang International Circuit	Kuala Lumpur	Malaysia	null	null
3	Bahrain International Circuit	Sakhir	Bahrain	Bahrain Grand Prix	2
4	Circuit de Barcelona-Catalunya	Montmeló	Spain	Spanish Grand Prix	5
5	Istanbul Park	Istanbul	Turkey	null	null
6	Circuit de Monaco	Monte-Carlo	Monaco	Monaco Grand Prix	6
7	Circuit Gilles Villeneuve	Montreal	Canada	Canadian Grand Prix	7
8	Circuit de Nevers Magny-Cours	Magny Cours	France	null	null
9	Silverstone Circuit	Silverstone	UK	British Grand Prix	10
10	Hockenheimring	Hockenheim	Germany	German Grand Prix	11
11	Hungaroring	Budapest	Hungary	Hungarian Grand Prix	12
12	Valencia Street Circuit	Valencia	Spain	null	null
13	Circuit de Spa-Francorchamps	Spa	Belgium	Belgian Grand Prix	13
14	Autodromo Nazionale di Monza	Monza	Italy	Italian Grand Prix	14
15	Marina Bay Street Circuit	Marina Bay	Singapore	Singapore Grand Prix	15

72 rows

Semi Join

It works as Inner Join but you will get data from the Left Table only

```
# semi_join
race_circuits_df = circuits_df.join(races_df, circuits_df.circuit_Id == races_df.circuit_id, "semi")
```

```
display(race_circuits_df)
```

Table

	circuit_Id	circuit_ref	circuit_name	location	country	latitude	longitude
1	1	albert_park	Albert Park Grand Prix Circuit	Melbourne	Australia	-37.8497	144.9
2	3	bahrain	Bahrain International Circuit	Sakhir	Bahrain	26.0325	50.51
3	4	catalunya	Circuit de Barcelona-Catalunya	Montmeló	Spain	41.57	2.261
4	6	monaco	Circuit de Monaco	Monte-Carlo	Monaco	43.7347	7.420
5	7	villeneuve	Circuit Gilles Villeneuve	Montreal	Canada	45.5	-73.52
6	9	silverstone	Silverstone Circuit	Silverstone	UK	52.0786	-1.016
7	10	hockenheimring	Hockenheimring	Hockenheim	Germany	49.3278	8.565
8	11	hungaroring	Hungaroring	Budapest	Hungary	47.5789	19.24
9	13	spa	Circuit de Spa-Francorchamps	Spa	Belgium	50.4372	5.971
10	14	monza	Autodromo Nazionale di Monza	Monza	Italy	45.6156	9.281
11	15	marina_bay	Marina Bay Street Circuit	Marina Bay	Singapore	1.2914	103.8
12	17	shanghai	Shanghai International Circuit	Shanghai	China	31.3389	121.
13	18	interlagos	Autódromo José Carlos Pace	São Paulo	Brazil	-23.7036	-46.69
14	22	suzuka	Suzuka Circuit	Suzuka	Japan	34.8431	136.5

18 rows

Anti Join

It will give everything from the left dataframe which is not found in the right dataframe

```
# anti_join
race_circuits_df = circuits_df.join(races_df, circuits_df.circuit_Id == races_df.circuit_id, "anti")
```

```
display(race_circuits_df)
```

Table

	circuit_Id	circuit_ref	circuit_name	location	country	latitude	lon
1	2	sepang	Sepang International Circuit	Kuala Lumpur	Malaysia	2.76083	
2	5	istanbul	Istanbul Park	Istanbul	Turkey	40.9517	
3	8	magny_cours	Circuit de Nevers Magny-Cours	Magny Cours	France	46.8642	
4	12	valencia	Valencia Street Circuit	Valencia	Spain	39.4589	
5	16	fuji	Fuji Speedway	Oyama	Japan	35.3717	
6	19	indianapolis	Indianapolis Motor Speedway	Indianapolis	USA	39.795	
7	20	nurburgring	Nürburgring	Nürburg	Germany	50.3356	
8	21	imola	Autodromo Enzo e Dino Ferrari	Imola	Italy	44.3439	
9	23	osterreichring	A1-Ring	Spielburg	Austria	47.2197	
10	25	galvez	Autódromo Juan y Oscar Gálvez	Buenos Aires	Argentina	-34.6943	
11	26	jerez	Circuito de Jerez	Jerez de la Frontera	Spain	36.7083	
12	27	estoril	Autódromo do Estoril	Estoril	Portugal	38.7506	
13	28	okayama	Okayama International Circuit	Okayama	Japan	34.915	
14	29	adelaide	Adelaide Street Circuit	Adelaide	Australia	-34.9272	

51 rows

Cross Join

```
# cross_join
race_circuits_df = races_df.crossJoin(circuits_df)
```

```
display(race_circuits_df)
```

Table

	race_id	race_year	round	circuit_id	race_name	ingestion_date	race_timestamp
1	1010	2019	1	1	Australian Grand Prix	2024-07-21T09:58:36.897+00:...	2019-03-17T05:10:0
2	1010	2019	1	1	Australian Grand Prix	2024-07-21T09:58:36.897+00:...	2019-03-17T05:10:0
3	1010	2019	1	1	Australian Grand Prix	2024-07-21T09:58:36.897+00:...	2019-03-17T05:10:0
4	1010	2019	1	1	Australian Grand Prix	2024-07-21T09:58:36.897+00:...	2019-03-17T05:10:0
5	1010	2019	1	1	Australian Grand Prix	2024-07-21T09:58:36.897+00:...	2019-03-17T05:10:0
6	1010	2019	1	1	Australian Grand Prix	2024-07-21T09:58:36.897+00:...	2019-03-17T05:10:0
7	1010	2019	1	1	Australian Grand Prix	2024-07-21T09:58:36.897+00:...	2019-03-17T05:10:0
8	1010	2019	1	1	Australian Grand Prix	2024-07-21T09:58:36.897+00:...	2019-03-17T05:10:0
9	1010	2019	1	1	Australian Grand Prix	2024-07-21T09:58:36.897+00:...	2019-03-17T05:10:0
10	1010	2019	1	1	Australian Grand Prix	2024-07-21T09:58:36.897+00:...	2019-03-17T05:10:0
11	1010	2019	1	1	Australian Grand Prix	2024-07-21T09:58:36.897+00:...	2019-03-17T05:10:0
12	1010	2019	1	1	Australian Grand Prix	2024-07-21T09:58:36.897+00:...	2019-03-17T05:10:0
13	1010	2019	1	1	Australian Grand Prix	2024-07-21T09:58:36.897+00:...	2019-03-17T05:10:0

14	1010	2019	1	1	Australian Grand Prix	2024-07-21T09:58:36.897+00:...	2019-03-17T05:10:0

1,449 rows

1449